

Project Lifecycle & Standard Operating Procedure

Development Template

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Introduction

This document outlines the standard workflow for initializing, developing, packaging, and distributing projects built with the standardized template. Following this procedure ensures consistency, scalability, and ease of maintenance across all personal projects.

Project Initialization

1. **Generate Template:** Run the `create_project.sh` script in the desired parent directory.
2. **Environment Setup:** Navigate to the new project root and execute `bash scripts/setup_env.sh`. This configures the Conda environment and installs dependencies defined in `requirements.txt` and `requirements-dev.txt`.

Development Workflow

- **Configuration:** Use `src/config.py` and `assets/themes/*.json` to manage UI themes and application settings.
- **File Handling:** Always use `src/core/utils.py`'s `get_resource_path()` function to reference assets (images, icons, etc.) to ensure compatibility when packaged as an executable.
- **Logic & UI:** Maintain a strict separation of concerns. Keep `ui/` and `core/` logic isolated, using `main.py` only as the entry point to wire components together.

Packaging (PyInstaller)

Before distributing, ensure the `build/` and `dist/` directories are clean.

1. **Linux:** Use `bash scripts/compile_ubuntu.sh`. Ensure the script includes `-add-data="assets:assets"` to bundle theme and image files.
2. **Windows:** Use `scripts/compile_windows.bat`. Ensure similar inclusion of assets using semi-colon syntax (`-add-data="assets;assets"`).

Distribution & Repository Maintenance

- **Version Control:** Always verify the `.gitignore` file before staging files. Ensure that compiled binaries (`.exe`, `.AppImage`), local databases (`*.db`, `*.sqlite`), and cache folders are ignored.
- **Documentation:** Keep the `README.md` updated with usage instructions specific to the project's current feature set.
- **Database Management:** Keep SQL schema blueprints in the `database/` folder to facilitate environment recreation.